MARIGANESH M

Biomedical Engineer

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SUMMARY

Enthusiastic Biomedical Engineering graduate with a strong foundation in medical technology and hands-on project experience. Skilled in biomedical system design, web development, and basic Python programming. Passionate about leveraging technology to improve patient care and support healthcare innovation. Adaptable, quick to learn, and eager to contribute to impactful engineering solutions.

EDUCATION

Sri muthukumaran Institute of technology
B.E Biomedical Engineering 2022 – 2026
CGPA - 7.9

Government Hr Sec School, Pattukottai
HSC - 55.3% 2021 - 2022
SSLC - 57.2% 2019 - 2020

SKILLS

Technical Skills	Tools
Embedded Systems	Microsoft Office
IoMT (Internet of Medical Things)	VS Code
Medical Instrumentation	Google Colab
Python (Basic)	solidworks (learning)
HTML, CSS, JavaScript, React	canva

CERTIFICATIONS

- Harvey Biomedical (Medical Device Quality Assurance)
- Deepam Academy Of Medical Sciences(3D Printing For Artificial Organs)
- International Conference MDDDS (Paper Presentation)
- Molecular Innovations In Entomology For Agricultural Sustainability
- IEEE-SBC61411A (Research Presentation)

LANGUAGES

- TAMIL
- ENGLISH

INTERESTS

Interested in embedded systems, IoMT, IoT, web design, and creative freelancing projects using Canva.

HOSPITAL TRAINING

Deepam Hospital, Chennai, India **Mar-2024**
Adequately prepared in ultrasound | ICU Equipment| healthcare Equipment | Physiotherapy Therapy Equipment

Chennai Meenakshi Multispeciality Hospital, Chennai **Jul-2025**
Hands-on troubleshooting of departmental equipment under the guidance of experienced biomedical engineers, with real-world application of biomedical principles in patient care and hospital operations.

INTERNSHIPS

Focus Imaging Systems **Oct 2024**
An internship at Focus Imaging Systems offers real-world experience with medical imaging equipment and helps build practical biomedical engineering skills

Neura Ai, Chennai **Oct-2025**
Designed and simulated a 660 nm LLLT (Low-Level Laser Therapy) Hair Growth Stimulator.

Phoenix Medical System **Mar - June 2026**
Worked on neonatal medical device assembly, SAP documentation, troubleshooting, fixing, and final testing in a medical manufacturing environment.

Ababil Healthcare Private Limited **July - October 2026**
Installs, calibrates, and repairs critical care medical equipment while managing regional maintenance schedules for Ababil Healthcare.

WORKSHOPS

- Hands-on experience in dismantling and reassembling syringes, infusion pumps, and ICU monitoring equipment to build strong technical knowledge and practical skills.

PROJECTS

Wireless Non-Invasive Approach for Blood Glucose Monitoring
- A wireless non-invasive blood glucose monitor lets people track their sugar levels comfortably without drawing blood. It uses smart sensors to read glucose levels through the skin and shares the data instantly with a mobile app or doctor.

EEG-Based Alzheimer's Disease Detection Using Machine Learning
- Developed an EEG-based Alzheimer's disease detection system using machine learning to classify patients into Healthy, Mild, Moderate, and Severe stages. Time-domain features were extracted from EEG signals, and models like Random Forest, KNN, and SVM were evaluated, with Random Forest achieving the best performance. A web application was also created to enable real-time prediction and visualization of EEG data.